Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 36)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Explain why organic pollution of rivers is usually monitored by measuring Biochemical Oxygen Demand (BOD) rather than the concentrations of the pollutants themselves.

[3 marks]

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0 1 . 2 Explain why a coliform count is used to assess the health risk from sewage contamination rather than testing for pathogens such as typhoid.

[2 marks]

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0 1 . 3 Give **three** characteristics of species that make them useful in a biotic index.

[3 marks]

1 .....

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2 .....

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3 .....

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0 1 . 4 Give **one** advantage and **one** disadvantage of using a biotic index rather than chemical monitoring.

[2 marks]

Do not write outside the box

Advantage .....

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Disadvantage .....

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0 2 A sand and gravel pit near a town is due to close. Three options for the site are proposed: a wetland nature reserve, housing, or return to farmland.

0 2 . 1 Explain why flooded sand and gravel pits are often developed as wetland nature reserves.

[2 marks]

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0 2 . 2 Explain the conditions that would need to be met for the site to be used for housing or farmland.

[2 marks]

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0 2 . 3 Explain why moving the wildlife from a site before mining begins is rarely completely successful.

[1 mark]

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0 3 Table 1 compares a 1000 MW coal-fired power station with a 1000 MW nuclear power station.

Table 1

Feature	Coal-fired	Nuclear (uranium)
Fuel used per year	3.5 million tonnes	30 tonnes
CO_2 emitted per kWh / g	900	12 (mostly fuel processing and construction)
Construction cost / £ billion	2	20
Fuel cost per kWh / pence	3.0	0.5
Typical response to changes in demand	Hours	Slow (days)

0 3 . 1 Explain how nuclear fission releases energy in a reactor.

[3 marks]

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0 3 . 2 Use Table 1 to explain why nuclear power is used for base-load electricity but not for meeting peaks in demand.

[2 marks]

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0 3 . 3 Use Table 1 to explain why nuclear power stations can be built where coal-fired stations could not.

[2 marks]

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0 3 . 4 Explain why the CO_2 emissions of nuclear power are not zero.

[1 mark]

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0 3 . 5 Give **two** reasons why the true cost of nuclear electricity is uncertain.

[2 marks]

1 .....

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2 .....

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0 4 Table 1 shows some radioactive isotopes, their half-lives and the type of radiation they emit.

Table 1

Isotope	Half-life	Main radiation	Source
Iodine-131	8 days	Beta / gamma	Fission in reactors
Strontium-90	29 years	Beta	Fission in reactors
Caesium-137	30 years	Beta / gamma	Fission in reactors
Plutonium-239	24 400 years	Alpha	Neutron bombardment of U-238
Uranium-238	4.5×10^9 years	Alpha	Occurs naturally in rocks

0 4 . 1 Define the term **half-life**.

[1 mark]

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0 4 . 2 A sample of iodine-131 has an activity of 6400 Bq. Calculate its activity after 32 days.

[2 marks]

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Do not write outside the box

Answer [ ] Bq

0 4 . 3 Use Table 1 to explain why isotopes with half-lives of several decades, such as caesium-137, cause the greatest concern after a nuclear accident.

[3 marks]

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0 4 . 4 Explain why plutonium-239 is relatively safe outside the body but very dangerous if inhaled or ingested.

[3 marks]

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0 4 . 5 Explain what is meant by an **activation product**.

[1 mark]

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0 5 . 1 Which of the following is the main advantage of composting organic waste rather than landfilling it? Tick **one** box.

[1 mark]

A Aerobic decomposition produces little methane and returns nutrients to the soil ☐

B It produces more electricity ☐

C It removes heavy metals ☐

D It destroys asbestos fibres ☐

0 5 . 2 Explain why the ash from waste incinerators must be disposed of carefully.

[2 marks]

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0 5 . 3 Give **two** features of a well-managed landfill site that reduce pollution.

[2 marks]

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0 6 A student used 0.5 m x 0.5 m grid quadrats to survey ground flora in two meadows, one grazed and one ungrazed. Table 1 shows the results for the grazed meadow (10 quadrats).

Table 1

Species	Number of individuals
Ribwort plantain	48
Perennial ryegrass	120
White clover	36
Daisy	24
Creeping buttercup	12

0 6 . 1 Explain why a 0.5 m x 0.5 m quadrat was chosen rather than a 10 cm x 10 cm or 100 m x 100 m quadrat.

[2 marks]

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0 6 . 2 Calculate Simpson's Index of Biodiversity (D) for the grazed meadow using D = N(N-1) / Σn(n-1). Show your working.

[3 marks]

Answer

0 6 . 3 The ungrazed meadow had D = 1.8 with the same number of species. Explain what this shows.

[2 marks]

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0 6 . 4 Explain why a point quadrat would be better than a frame quadrat in tall vegetation.

[2 marks]

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0 6 . 5 Give **one** limitation of using the DAFOR abundance scale.

[1 mark]

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0 7 A rural community in Pakistan is considering a low-head helical (Archimedes screw) turbine on a river rather than joining a high-head dam scheme.

0 7 . 1 Explain why low-head turbines are more suitable than high-head HEP schemes for many rural areas.

[5 marks]

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0 7 . 2 Suggest **one** reason why such schemes may not be installed even at suitable sites.

[1 mark]

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0 7 . 3 The maximum flow rate at the site is 6.5 $m^3 s^{-1}$ giving an output of 100 kW at 90% turbine efficiency. At 10% of maximum flow the efficiency falls to 40%. Calculate the output at 10% of maximum flow.

[2 marks]

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Answer [ ] kW

0 7 . 4 Outline **two** ways a high-head reservoir can affect the river downstream.

[2 marks]

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0 8 Table 1 shows the mean annual soil loss measured on four fields of one farm. Soil forms on this farm at about 1 t ha^{-1} yr^{-1}.

Table 1

Field	Slope	Land use	Mean annual soil loss / t ha^{-1} yr^{-1}
A	2%	Permanent grassland	0.8
B	9%	Maize, ploughed up and down the slope	24.0
C	9%	Maize, contour ploughed	9.0
D	1%	Spring barley, soil left bare over winter	6.5

0 8 . 1 Use Table 1 to identify the fields that are suffering **accelerated** soil erosion, and explain your answer.

[2 marks]

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0 8 . 2 Calculate the percentage reduction in soil loss achieved by changing field B to contour ploughing.

[2 marks]

Answer ☐ %

0 8 . 3 Explain how each of the following reduces erosion by water.

[3 marks]

Contour ploughing .....

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Terracing .....

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Rows of stones .....

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0 8 . 4 Field D is exposed to strong winds. Explain **four** methods, other than planting hedgerows, that would reduce wind erosion there.

[4 marks]

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0 8 . 5 Describe **four** impacts of the eroded soil **away from** the fields where it is lost.

[4 marks]

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0 9 A dam and reservoir are to be built on a river to supply water and regulate flow.

0 9 . 1 Explain **three** effects the reservoir may have on wildlife in and around the river.

[6 marks]

0 9 . 2 Explain why decomposition of flooded vegetation in a new reservoir contributes to climate change.

[2 marks]

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0 9 . 3 Explain how a river-regulation reservoir reduces both flooding and water shortages downstream.

[2 marks]

10

1 0 . 1 Discuss how the features of energy resources determine their suitability for different uses and their environmental sustainability.

[9 marks]

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Extra space

1 0 . 2 State **one** reason why nuclear power stations can be located where transport infrastructure is poor.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.

[25 marks]

OR

1 1 . 2 Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS